

# GST-Ready Accounting Ledger

A clean-room rebuild for Indian wholesale accounting — built around balance, auditability, GST correctness and explainability.

DBG-274

Ctrl\_alt\_Delivers

## CORE ACCOUNTING

1 **GST Engine**  
CGST + SGST  
or IGST

2 **Ledger Guard**  
Debit = Credit  
before commit

3 **Immutable Void**  
Reverse, don't  
delete history

4 **AI Copilot**  
Read-only  
financial answers

DESIGNED FROM V2 PRD / GAPS / ARCHITECTURE / API / DATA MODEL

# Bigcapital's accounting core leaves four high-impact gaps.

The rebuild focuses only on the hard accounting path — not the full Bigcapital feature set.

GAP 01

CRITICAL

## Unbalanced journal writes

No pre-commit check that  $\Sigma \text{Debit} = \Sigma \text{Credit}$ .

GAP 02

CRITICAL

## No Indian GST split

Tax is modeled as one scalar + generic tax liability.

GAP 03

HIGH

## Void deletes history

Invoice deletion physically removes ledger rows.

GAP 07/08

HIGH

## Retry + atomicity risk

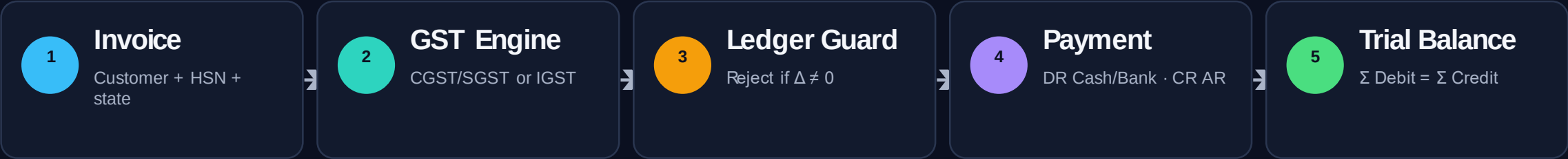
No idempotency; ledger work can span non-atomic queue writes.

REBUILD PRINCIPLE

Deterministic accounting engine → every write is balanced → every reversal is auditable.

# One deterministic accounting path from invoice to proof.

Core GST, double-entry enforcement, settlement and Trial Balance are designed as one transaction boundary.



EXAMPLE — ₹1,00,000 INTRA-STATE SALE @ 18%

|                     |        |           |
|---------------------|--------|-----------|
| Accounts Receivable | DEBIT  | ₹1,18,000 |
| Sales Income        | CREDIT | ₹1,00,000 |
| Output CGST Payable | CREDIT | ₹9,000    |
| Output SGST Payable | CREDIT | ₹9,000    |

Balanced journal: ₹1,18,000 debit = ₹1,18,000 credit

MUST-HAVE CORE

- Dual GST: CGST + SGST / IGST
- Immutable invoice reversal
- Live Trial Balance
- Seeded Chart of Accounts

Source: PRD.md + ARCHITECTURE.md + API.md

# AI Accounting Copilot — explain the books without controlling them.

Natural-language answers are grounded in live accounting records and kept strictly read-only.

## AI COPILOT

Why is invoice INV-0001 ₹11,800?

### Grounded answer

Taxable value: ₹10,000  
CGST 9%: ₹900 + SGST 9%: ₹900  
Total: ₹11,800

Source: INV-0001 · GST split · ledger records

## STRICT READ-ONLY BOUNDARY

AI CAN

Explain invoices, GST, payments, journal entries, trial balance

AI CANNOT

Create/modify invoices, payments or journals

AI CANNOT

Change tax values, delete records, bypass authorization

ENGINE

Deterministic ledger remains the source of truth

# Three Killer Tests define success.

Build the accounting core first; add AI only after the ledger is demonstrably correct.

01

## Every transaction balances

Reject any write when  $|\Sigma\text{Debit} - \Sigma\text{Credit}| \geq 0.0001$ .

HTTP 422 · 0 DB rows

02

## Void = reversal, not deletion

Keep original journal lines; append exact contra-entries.

Status **VOIDED** · audit preserved

03

## 20 random operations stay balanced

Mix invoices, partial/full payments and voids.

Trial Balance difference = 0

### IMPLEMENTATION ORDER

Core GL

KT-1

KT-2

KT-3

Fix 1

AI Copilot

AI tests

Re-run KT-1/2/3

Regression

**SUCCESS = balanced · reversible · auditable · GST-aware · explainable**